Hospital X
SAMueL Analysis
(Stroke Audit Machine Learning)

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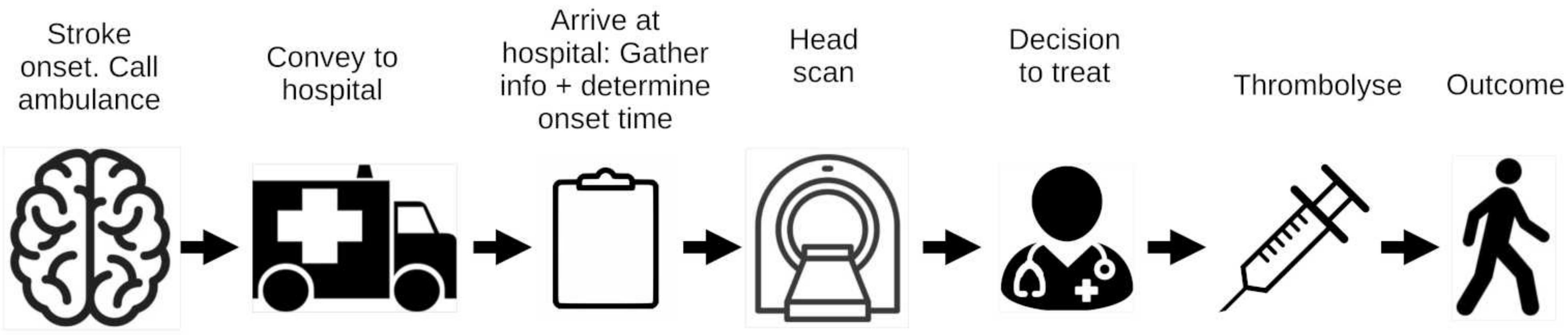
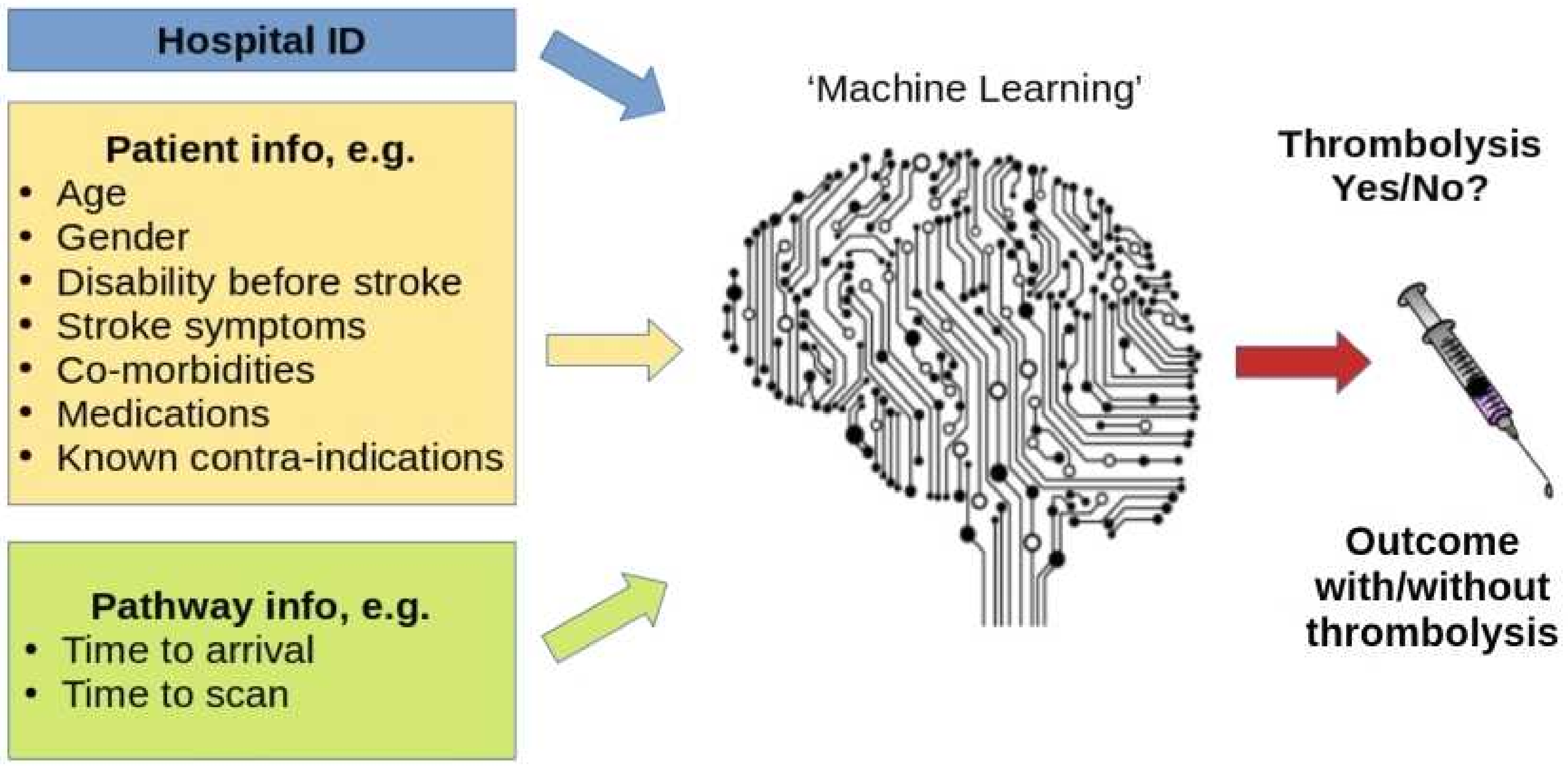
Breaking down the emergency stroke pathway into key steps



- We can model key changes to pathway, such as:
  - What if the pathway were **faster**?
  - What if hospital **determined the stroke onset time** in more patients?
  - What if **clinical decision-making** was like that of benchmark hospitals?
    - Predict what treatment a patient would receive at other hospitals
- We model these changes with a hospital’s own patient population, to allow for inter-hospital variation in patient population characteristics.

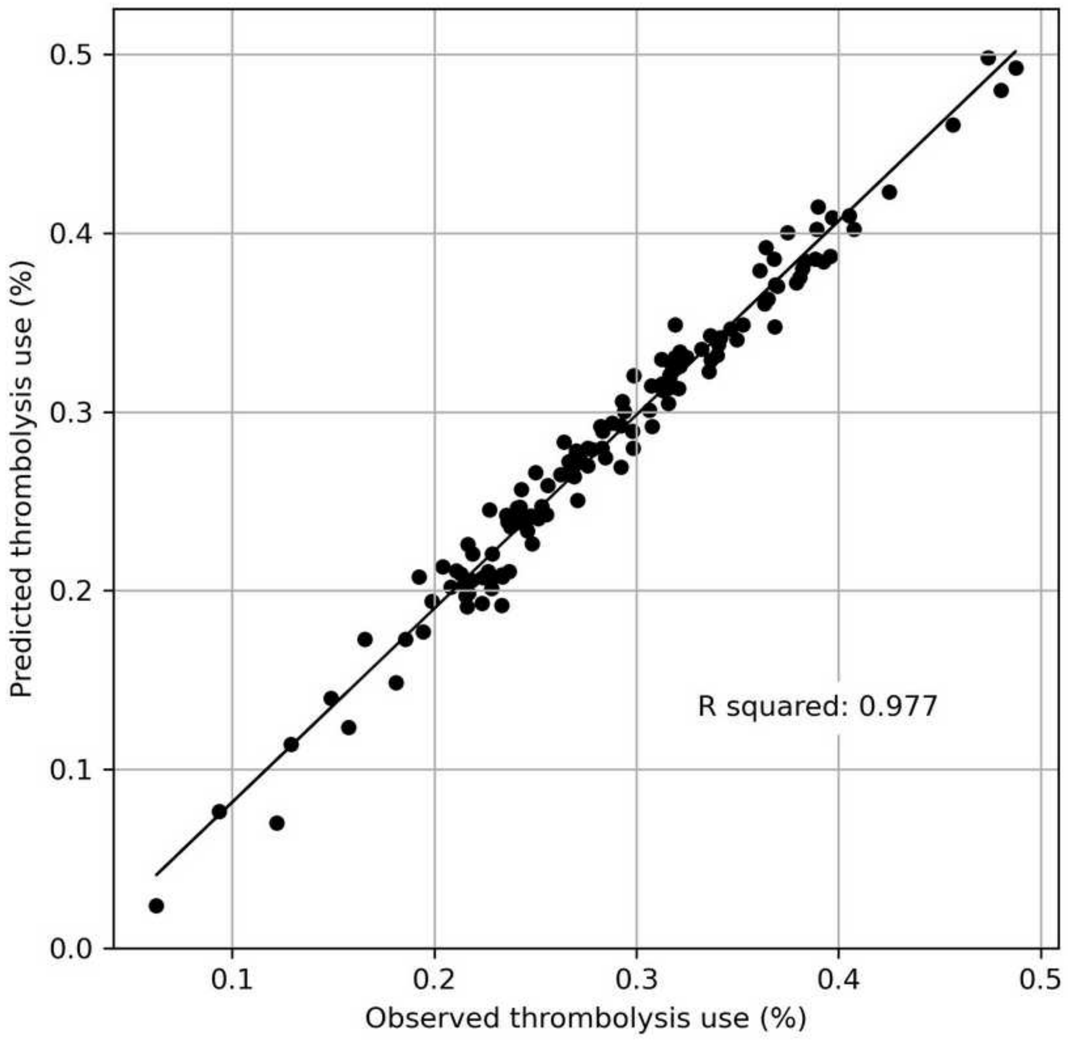
Machine learning overview



- Machine learning is based on the simple principle of recognising similarity to what has been seen before

- We accessed 246,676 emergency stroke admissions in England and Wales over three years. Our machine learning models use XGBoost classification, and are based on those 88,928 patients who arrive within 4 hours of known stroke onset. Accuracy = 85% (ROC AUC = 0.92).

Predicting hospital thrombolysis use across 132 hospitals



- Predicted thrombolysis use (for patients arriving within 4 hours of known stroke onset) was measured by combining 5 test sets from k-fold validation (where the model is tested on data not used to train the model).
- The predicted vs. observed thrombolysis are highly correlated (r-squared 0.977).

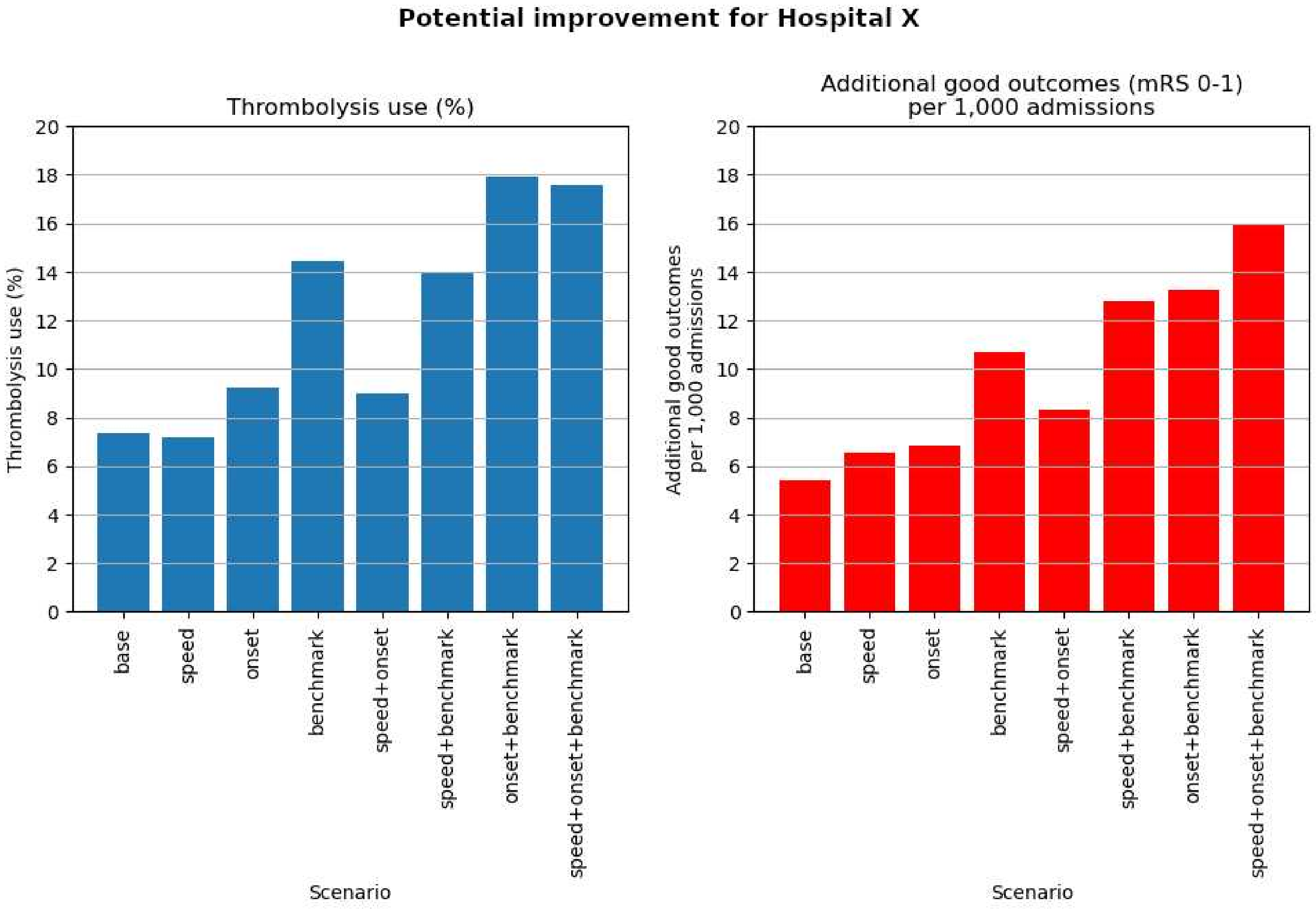
Descriptive stats

- Stats are for 5 years (2017-2021)
- Out of hospital admissions only
- General observations:
  - 60% of patients have stroke time determined (lower than national average)
- For patients arriving within 4 hours of known stroke onset
  - Age is older than national average
  - Prior disability is similar to national average
  - Stroke severity is a *little* lower than national average
  - Ambulance response, on-scene, and travel times are similar to national average (2017-2021)
  - Arrival-to-scan time is faster than national average
  - Scan-to-thrombolysis time is slower than national average
    - Distances?
    - Time for decision-making?
    - Other

Statistics for patients arriving within 4 hours of known stroke onset

	All teams 1Q	All teams median	All teams 3Q	Hospital X
admissions	138.00	183.00	239.00	172.00
age	73.30	74.99	75.98	77.68
age > 80	0.37	0.41	0.45	0.51
male	0.51	0.53	0.54	0.48
prior_disability	0.87	1.02	1.20	1.02
prior_disability_0-2	0.77	0.81	0.83	0.79
stroke_severity	8.25	9.11	9.63	8.25
onset_known	1.00	1.00	1.00	1.00
onset_to_arrival_time	98.00	104.00	110.00	102.00
onset_within_4hrs	1.00	1.00	1.00	1.00
precise_onset_known	0.53	0.65	0.73	0.72
onset_during_sleep	0.01	0.03	0.06	0.03
arrive_by_ambulance	0.88	0.91	0.93	0.92
call_to_ambulance_arrival_time	15.62	18.00	21.00	16.00
ambulance_on_scene_time	25.62	28.00	30.50	28.00
ambulance_travel_to_hospital_time	14.00	16.00	20.00	19.00
arrival_to_scan_time	22.00	28.00	35.00	22.00
thrombolysis	0.25	0.28	0.33	0.19
scan_to_thrombolysis_time	27.00	34.00	40.00	43.00
discharge_disability	2.61	2.85	3.04	3.08
discharge_disability_0-2	0.42	0.49	0.54	0.40
discharge_disability_5-6	0.21	0.23	0.26	0.27

Improvement potential (modelled on own hospital's patients)



- Improving speed to 30 minutes arrival-to-thrombolysis has only a small effect on thrombolysis use, but will improve net benefit (as majority of patients benefit).
- Determination of more stroke onset times is likely to lead to some improvement in thrombolysis use
- Decision to use thrombolysis in patients arriving within 4 hours of known stroke onset is significantly lower than benchmark hospitals (see next slides).

Comparing thrombolysis decision to '*benchmark*' hospitals

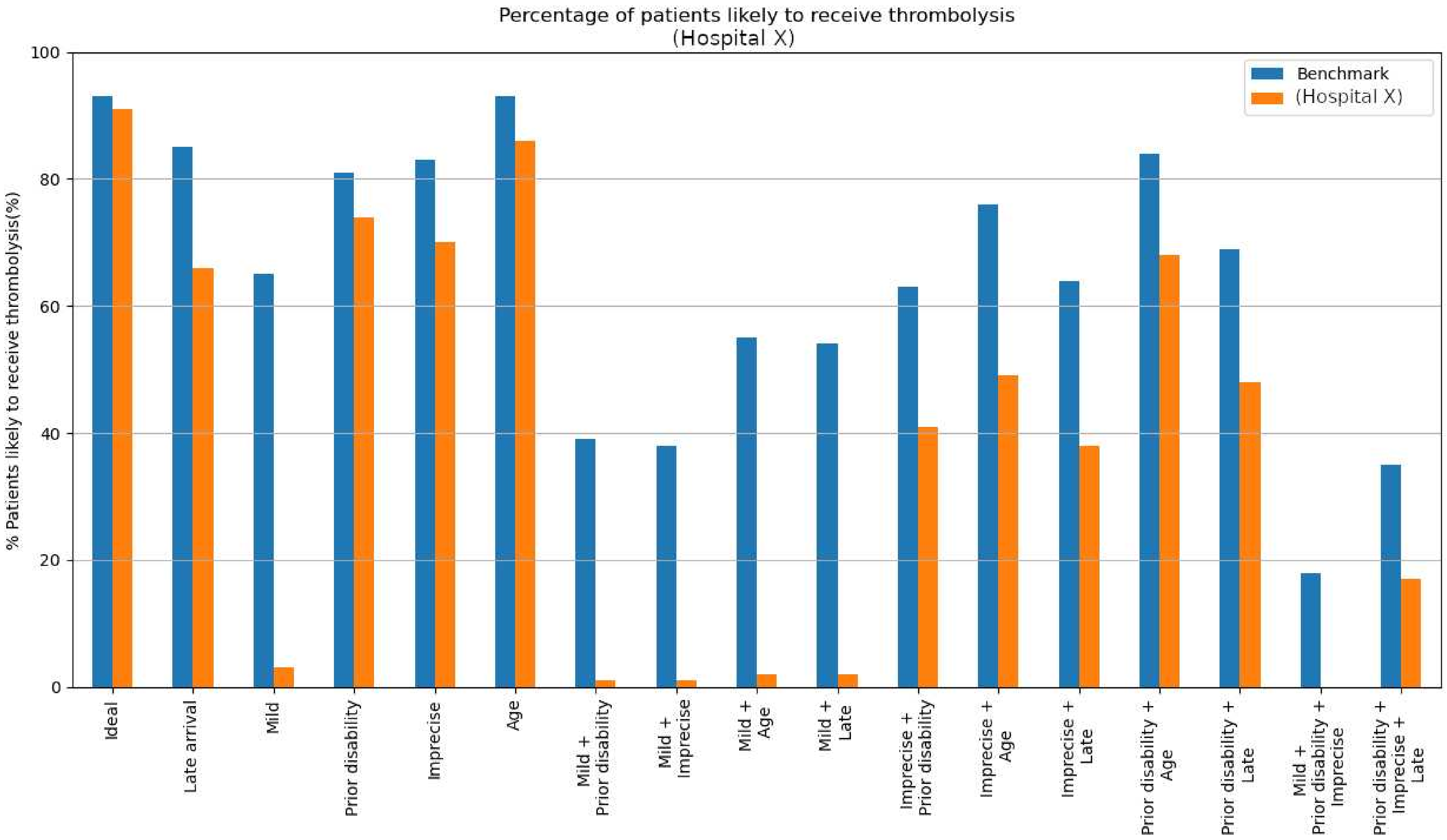
Benchmark hospitals are the 25 hospitals in England and Wales who are most likely to use thrombolysis if all hospitals saw the same patients (through to time of scan)

An *ideally* thrombolysable patient:

- Infarction
- 90 mins onset to arrival
- Onset not during sleep
- Precise onset time
- NIHSS 15
- No prior disability
- No use of anticoagulants for atrial fibrillation
- Age 72

Non-ideal changes:

- *Late*: Arrival at 3hr 45 min after stroke
- *Mild stroke*: NIHSS 3
- *Prior disability*: mRS 3 before stroke
- *Imprecise*: Imprecisely known stroke onset time
- *Age*: Age 87


Percentage of patients likely to receive thrombolysis
(Hospital X)
% Patients likely to receive thrombolysis(%)
Benchmark
(Hospital X)
0
20
40
60
80
100
Ideal
Late arrival
Mild
Prior disability
Imprecise
Age
Mild + Prior disability
Mild + Imprecise
Mild + Age
Mild + Late
Imprecise + Prior disability
Imprecise + Age
Imprecise + Late
Prior disability + Age
Prior disability + Late
Mild + Prior disability + Imprecise
Prior disability + Imprecise + Late

Predicted (learned) outcome for a mild stroke patient with or without thrombolysis

PATIENT CHARACTERISTICS

infarction: 1
prior_disability: 0
stroke_severity: 4
stroke_team: Hospital X
onset_to_arrival_time: 120
arrival_to_scan_time: 15
precise_onset_known: 0
onset_during_sleep: 0
onset_to_thrombolysis: 150
age: 75
any_afib_diagnosis: 0
afib_anticoagulant: 0

THROMBOLYSIS CHOICE

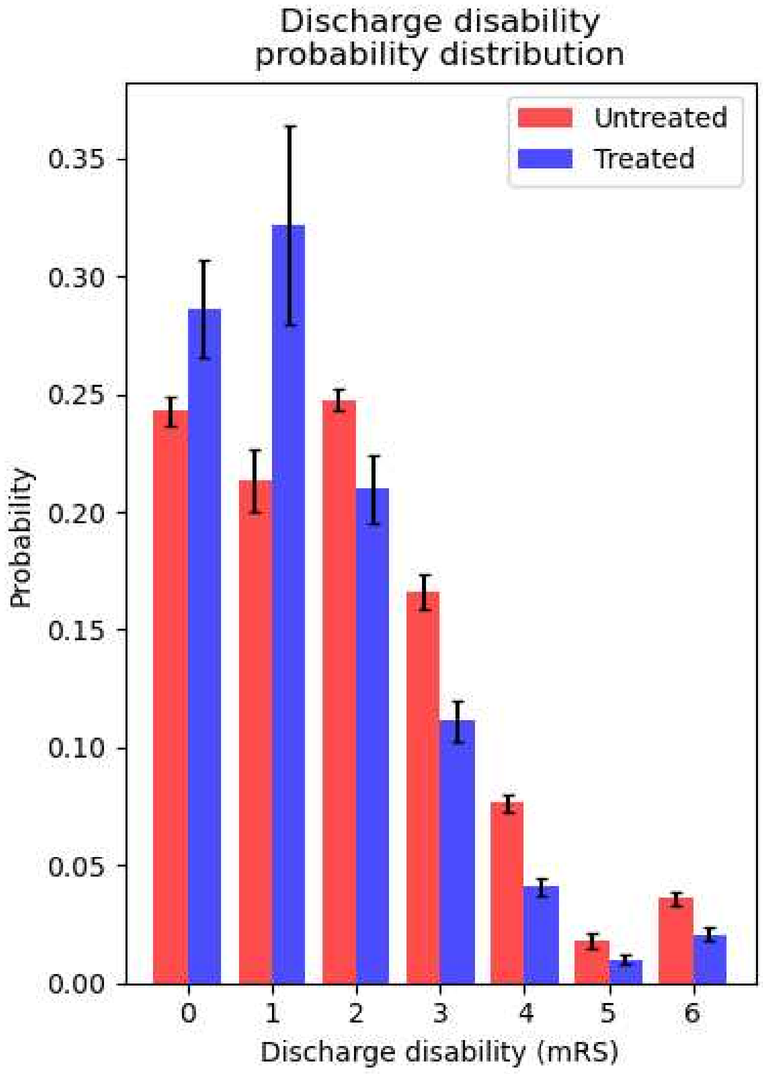
Model prediction = 0.06 (0.01)
Benchmark prediction = 0.59

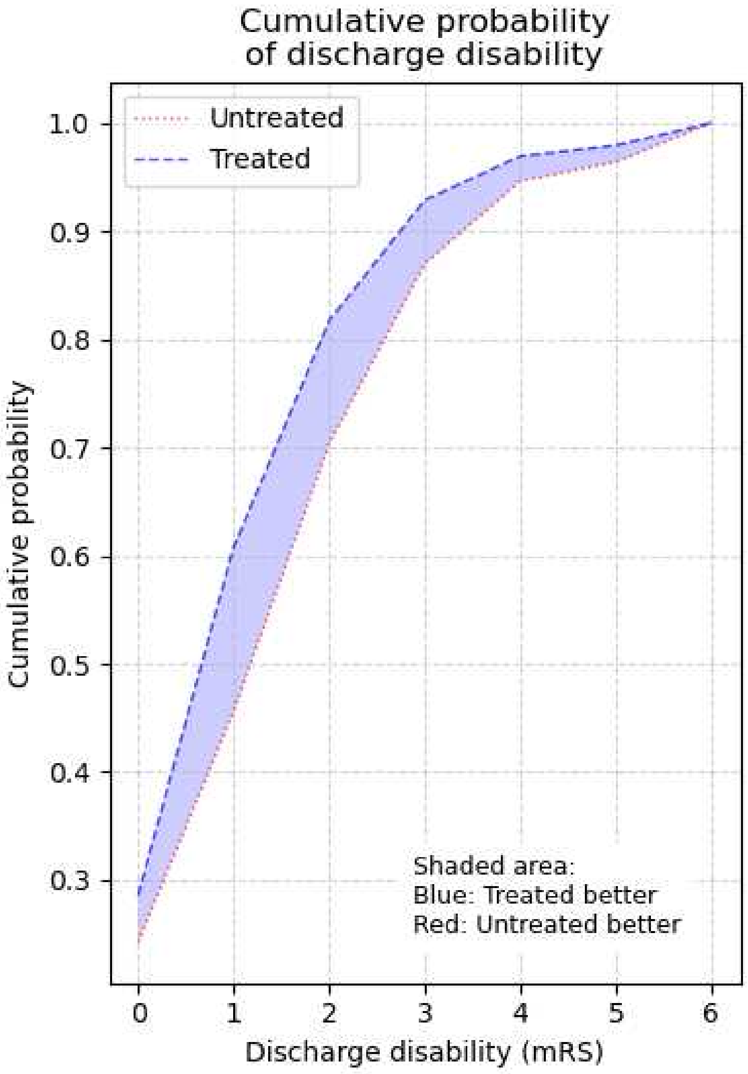
LIKELY OUTCOME

Untreated weighted mRS = 1.82 (0.03)
Treated weighted mRS = 1.41 (0.04)
mRS improvement = 0.41 (0.05)

Untreated proportion mRS 0-2 = 0.70 (0.01)
Treated proportion mRS 0-2 = 0.82 (0.01)
Change in proportion mRS 0-2 = 0.11 (0.00)

Untreated proportion mRS 5-6 = 0.05 (0.00)
Treated proportion mRS 5-6 = 0.03 (0.00)
Change in proportion mRS 5-6 = -0.02 (0.00)





Key observations

- Hospital X:
  - Determines stroke onset times in fewer patients than the national average
  - Has good arrival-to-scan times, but slower scan-to-thrombolysis times than the national average
  - Is particularly less likely to give thrombolysis to mild stroke patients compared to higher thrombolysing 'benchmark' hospitals.